

Trainings ▾

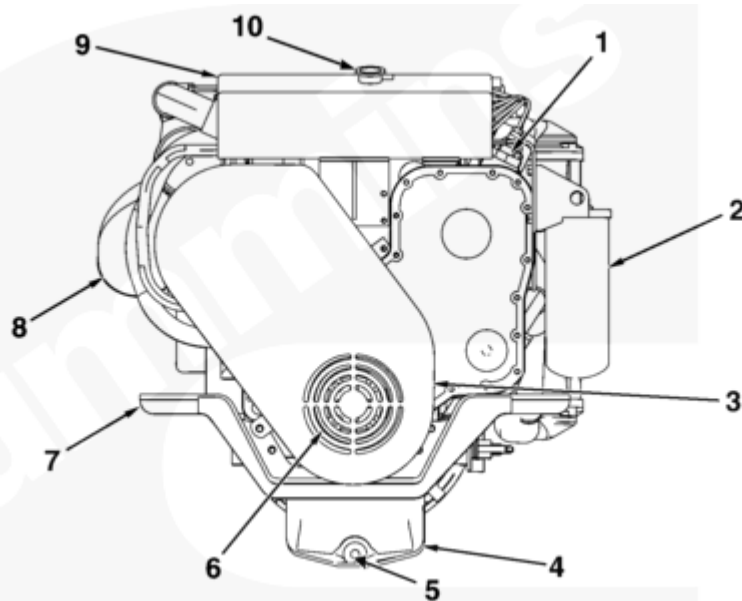
Engine Views

Marine Applications

The following drawings illustrate the major components. Maintenance personnel **must** be familiar with each component to conduct the maintenance and service procedures discussed in this manual.

Note : The drawings in this section reflect a standard configuration. The engine can be equipped with optional accessories such as port-or starboard-side dipsticks. Location of optional items can be different from those shown.

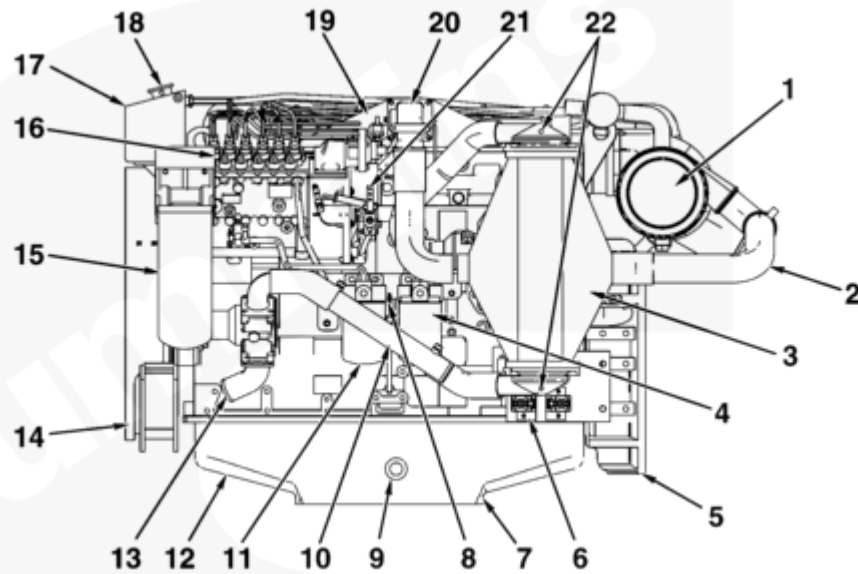
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00900315

Front View, C Series, Six-Cylinder Sea Water Aftercooled 450C

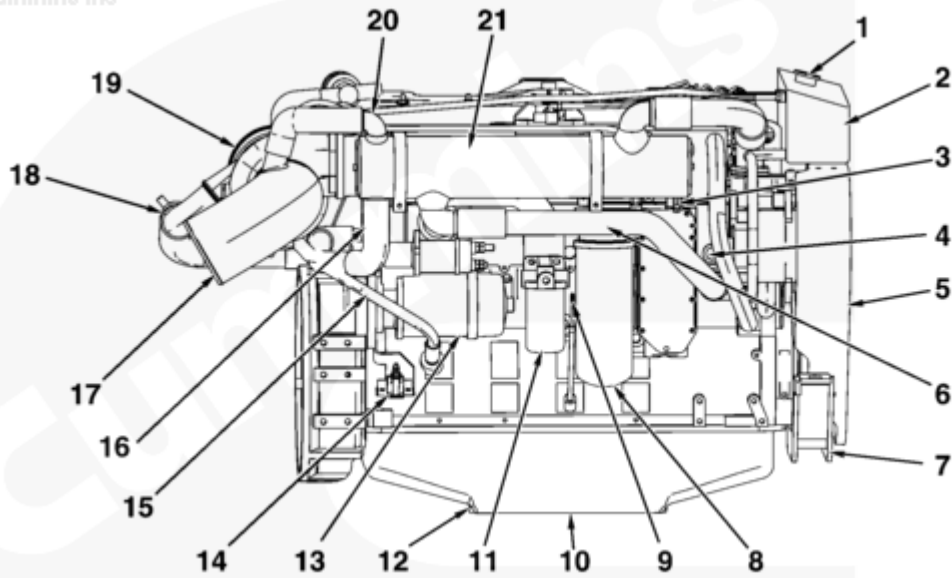
1. Fuel pump
2. Remote mounted oil filter (optional)
3. Belt guard
4. Oil pan
5. Engine oil drain, front
6. Vibration damper (behind cover)
7. Engine mount, front
8. Exhaust elbow
9. Expansion tank
10. Coolant fill.



00900312

Port View, C 8.3, Six-Cylinder Sea Water Aftercooled 450C

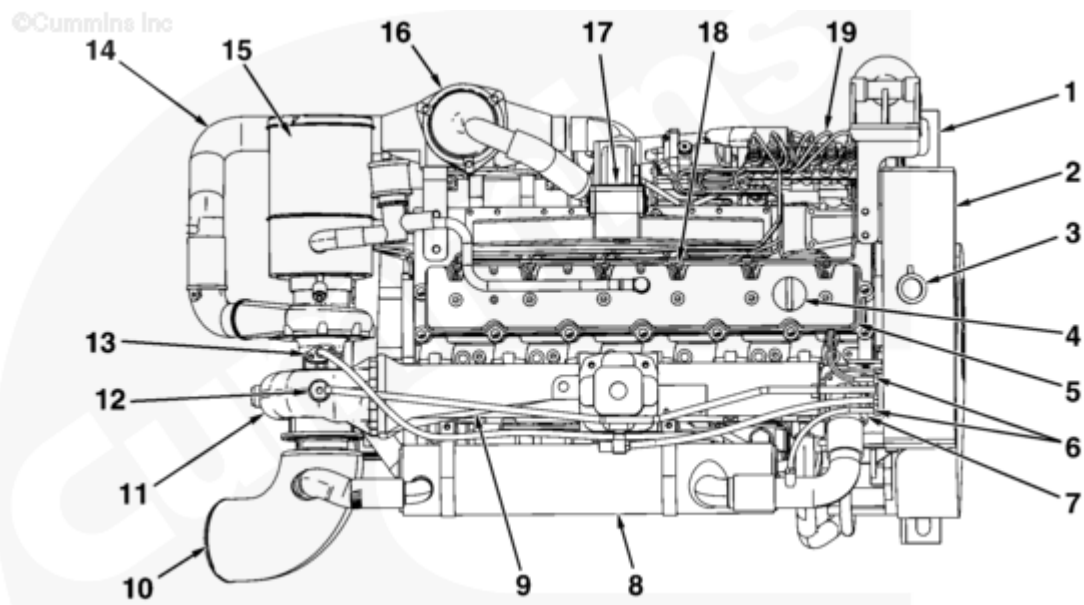
1. Air cleaner
2. Inlet air crossover tube
3. Aftercooler (sea water-type)
4. Water (coolant) filter
5. Flywheel housing
6. Air heater control solenoids
7. Engine oil drain, rear
8. Dipstick (engine oil) port-side standard
9. Oil pan heater location (if equipped)
10. Fuel cooler
11. Fuel filter (standard)
12. Oil pan
13. Sea water pump inlet (rotatable)
14. Belt guard
15. Remote mounted oil filter (optional)
16. Fuel pump
17. Expansion tank
18. Coolant fill
19. Stop lever
20. Air heater
21. Throttle lever
22. Zinc plugs (aftercooler).



00900313

Starboard View, C 8.3, Six-Cylinder Sea Water Aftercooled 450C

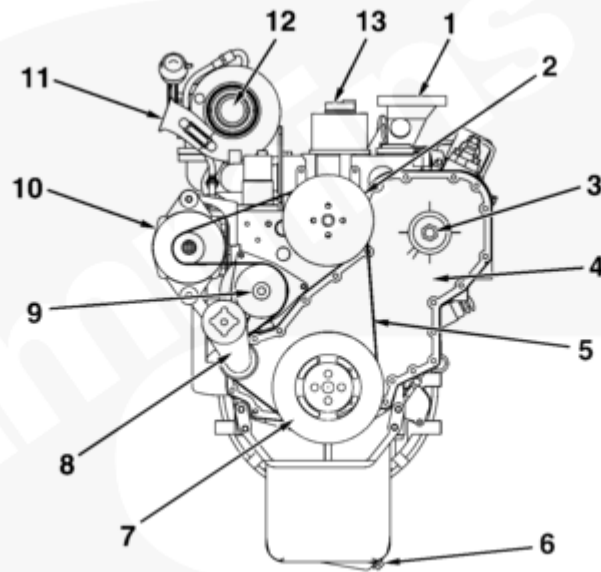
1. Coolant fill
2. Expansion tank
3. Zinc plug (heat exchanger)
4. Coolant temperature sensor (coolant drain)
5. Belt guard
6. Heat exchanger coolant outlet tube
7. Engine mount (front)
8. Engine oil filter
9. Engine oil dipstick (optional starboard-side)
10. Oil pan
11. Fuel filter (optional starboard-side)
12. Engine oil drain, rear
13. Starter
14. Starter auxiliary magnetic switch
15. Turbocharger oil drain tube
16. Sea water inlet to heat exchanger
17. Exhaust outlet elbow
18. Inlet air crossover tube
19. Turbocharger (water cooled)
20. Sea water outlet to exhaust
21. Heat exchanger.



Top View, C Series, Six-Cylinder Sea Water Aftercooled 450C

1. Engine mount (front)
2. Expansion tank
3. Coolant fill
4. Engine oil fill
5. Valve cover
6. Coolant vent line connections
7. Coolant thermostat location
8. Heat exchanger
9. Exhaust manifold (water cooled)
10. Exhaust outlet elbow
11. Turbocharger (water cooled)
12. Turbocharger coolant vent line
13. Turbocharger oil inlet
14. Inlet air crossover tube
15. Air filter
16. Aftercooler (sea water-type)
17. Air heater
18. Fuel injector
19. Fuel pump.

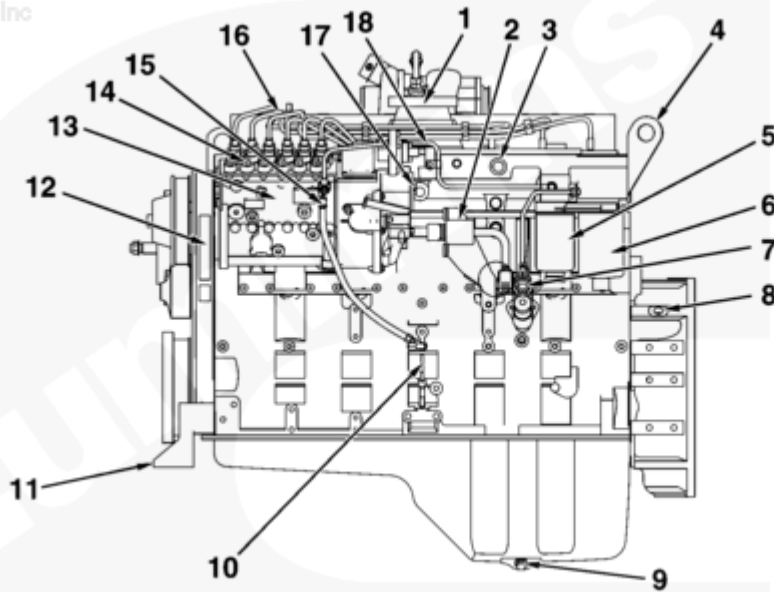
Industrial Applications



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Front View - C8.3 - Industrial

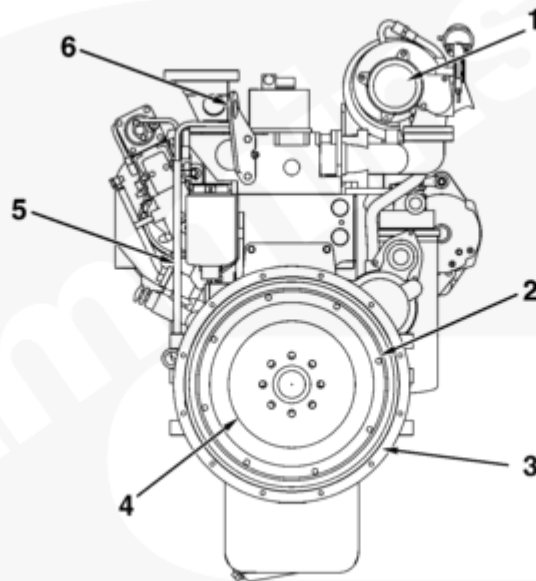
1. Engine air inlet
2. Fan pulley
3. Fuel pump drive cover
4. Front gear cover
5. Drive belt
6. Lubricating oil pan drain plug
7. Vibration damper
8. Fan belt tensioner
9. Water pump
10. Alternator
11. Turbocharger air outlet
12. Turbocharger air inlet
13. Engine oil fill.



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Left View - C8.3 - Industrial

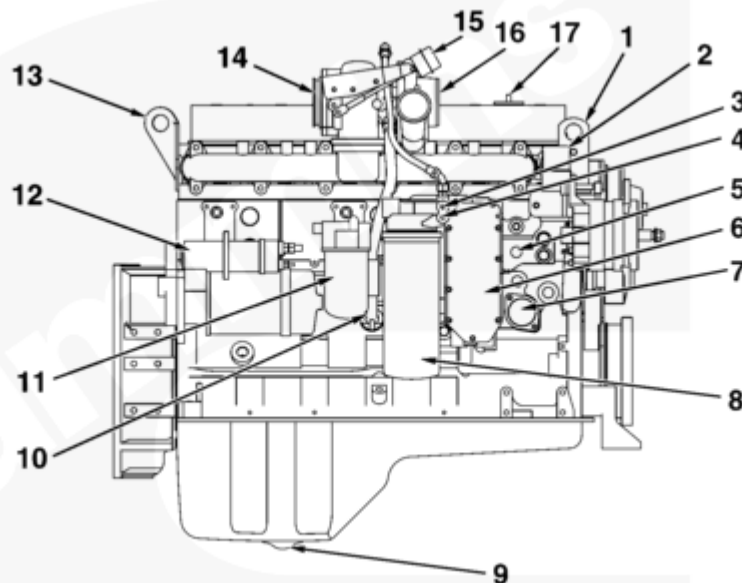
1. Engine air inlet
2. Shutoff solenoid
3. M22 x 1.5 inlet (air)
4. Rear lifting bracket
5. Fuel filter
6. Fuel/water separator
7. Fuel transfer pump
8. 3/4 x 16-inch UNF tap for magnetic pickup
9. Lubricating oil pan drain plug
10. Lubricating oil dipstick
11. Front engine mounting bracket
12. Engine dataplate
13. Fuel injection pump
14. Delivery valve
15. Fuel drain line
16. High-pressure fuel lines
17. 1/2-inch NPTF inlet (air)
18. Intake air heater.



00900321

Rear View - C8.3 - Industrial

1. Turbocharger exhaust outlet
2. Flexplate mounting holes
3. Flywheel housing
4. Flywheel
5. Crankcase breather tube
6. Rear engine lifting bracket.



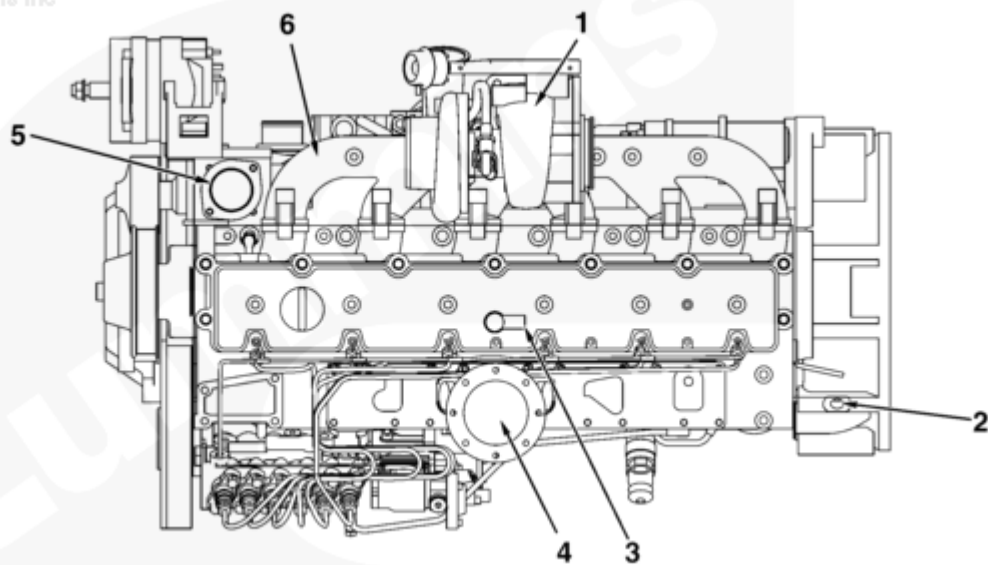
00900323

Right View - C8.3 - Industrial

1. Front engine lifting bracket
2. Water/coolant outlet connection
3. Lubricating oil pressure (after filter)
4. Lubricating oil pressure (before filter)
5. Coolant heater port
6. Lubricating oil cooler

7. Water/coolant inlet
8. Lubricating oil filter
9. Lubricating oil pan drain plug
10. Turbocharger oil drain
11. Coolant filter
12. Starting motor
13. Rear engine lifting bracket
14. Turbocharger exhaust outlet
15. Turbocharger wastegate actuator
16. Turbocharger air inlet
17. Engine oil fill.

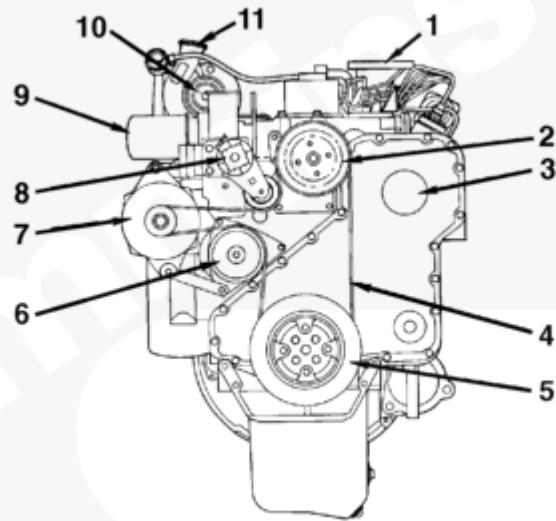
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00900324

Top View - C8.3 - Industrial

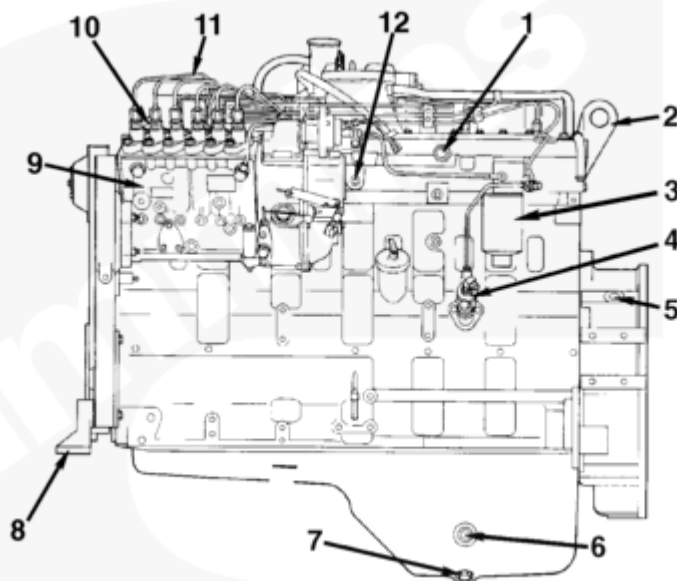
1. Turbocharger
2. 3/4 x 16-inch UNF tap for magnetic pickup
3. Crankcase breather
4. Engine air inlet
5. Thermostat
6. Exhaust manifold.



18900030

Front View - C8.3 - Automotive

1. Engine air inlet
2. Fan pulley
3. Fuel pump drive cover
4. Drive belt
5. Vibration damper
6. Water pump
7. Alternator
8. Belt tensioner
9. Coolant filter
10. Turbocharger air inlet
11. Turbocharger air outlet.



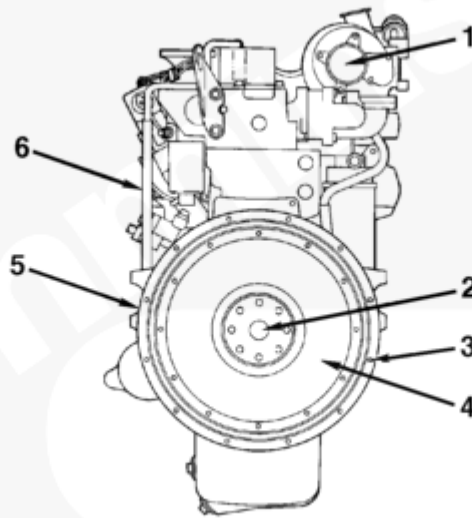
18900027

Left View - C8.3 - Automotive

1. M22 X 1.50 intake (air)

2. Rear lifting bracket
3. Fuel filter/water separator
4. Fuel transfer pump
5. 3/4 X 16-inch UNF tap for magnetic pickup
6. Provisions for oil heater
7. Lubricating oil drain plug
8. Front engine mounting bracket
9. Fuel injection pump
10. Distribution valve
11. High pressure fuel lines
12. 1/4-inch NPTF intake (air).

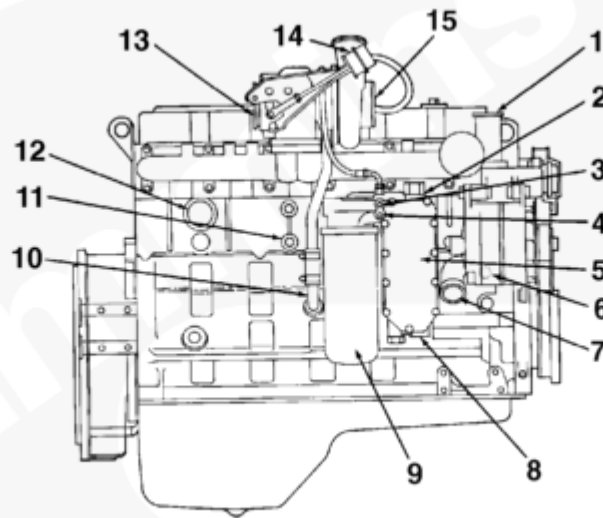
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18900029

Rear View - C8.3 - Automotive

1. Turbocharger exhaust outlet
2. Pilot bearing bore
3. Transmission mounting holes
4. Flywheel
5. Flywheel housing
6. Crankcase breather vent tube.



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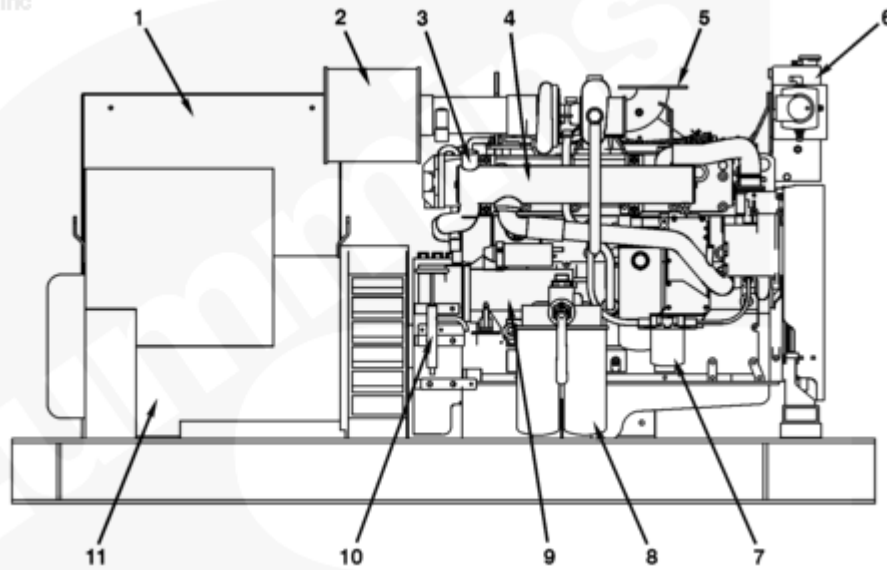
Right View - C8.3 - Automotive

1. Water outlet connection
2. Lubricating oil temperature thermostat
3. Lubricating oil pressure (after filter)
4. Lubricating oil pressure (before filter)
5. Lubricating oil cooler
6. 1/2-inch NPFT (coolant)
7. Coolant inlet
8. Lubricating oil temperature sensor
9. Lubricating oil filter
10. Turbocharger oil drain
11. Provision for cab heater
12. Provision for coolant heater
13. Turbocharger exhaust outlet
14. Turbocharger wastegate actuator
15. Turbocharger air inlet.

Generator Set Applications

The following drawings illustrate the major components. Maintenance personnel **must** be familiar with each component to conduct the maintenance and service procedures discussed in this manual.

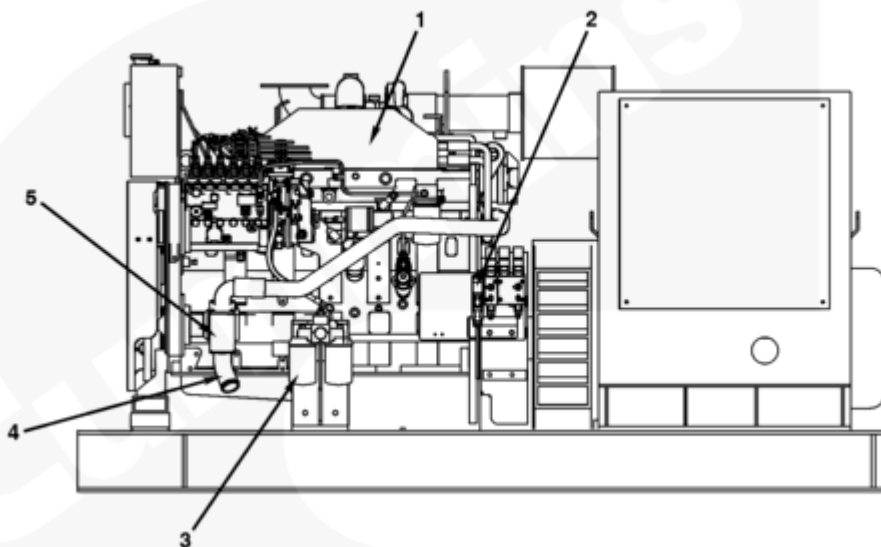
Note : The drawings in this section reflect a standard configuration. The engine can be equipped with optional accessories such as port-or starboard-side dipsticks. Location of optional items can be different from those shown.



00900423

Right View - C Series - C Power Heat Exchanger Cooled

1. Local control panel
2. Air cleaner
3. Sea water outlet connection
4. Heat exchanger
5. Exhaust connection
6. Coolant expansion tank
7. Coolant filter
8. Duplex lubrication oil filters
9. Starter
10. Lubricating oil sump hand pump
11. Alternator.



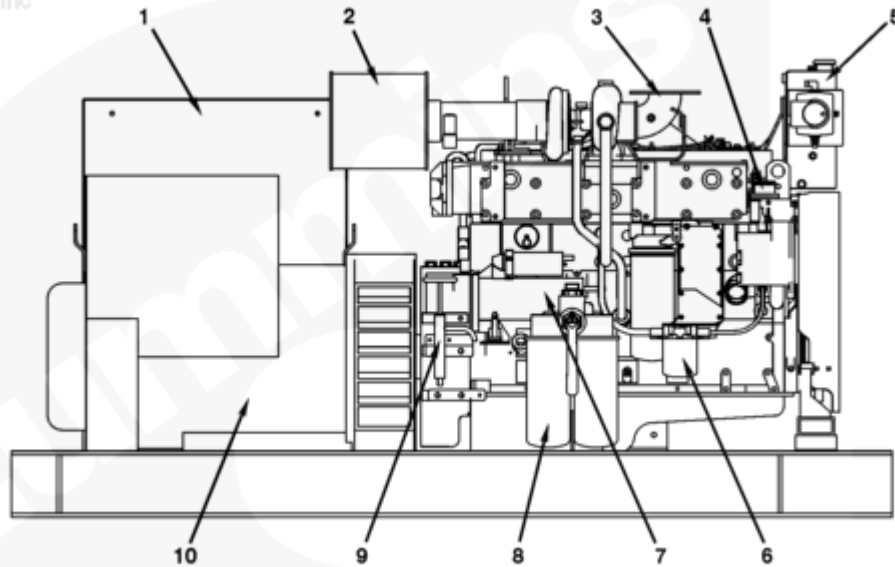
00900424

Left View - C8.3 - C Power Heat Exchanger Cooled

1. Aftercooler

2. Emergency shutdown sensors
3. Duplex fuel filters
4. Seawater inlet connection
5. Seawater pump

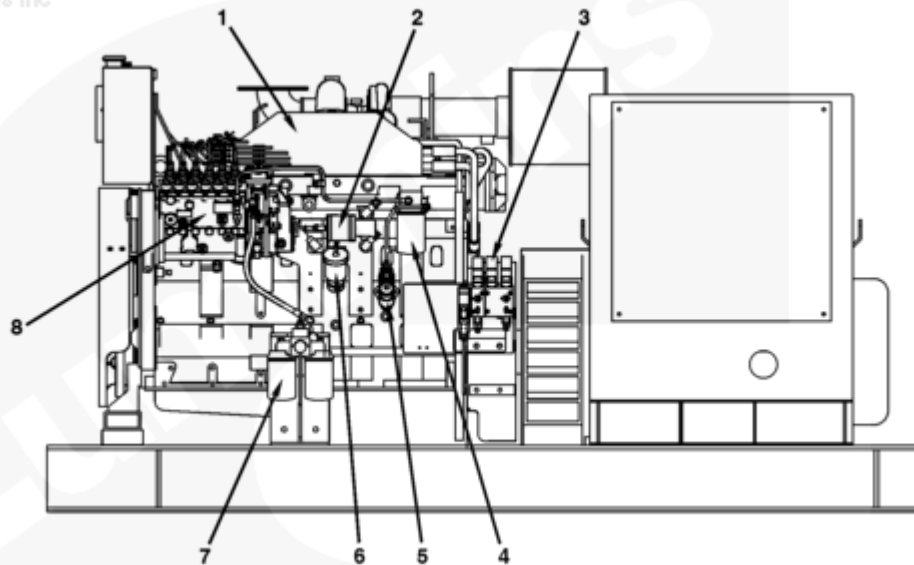
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00900425

Right View - C 8.3 - C Power Keel cooler Cooled

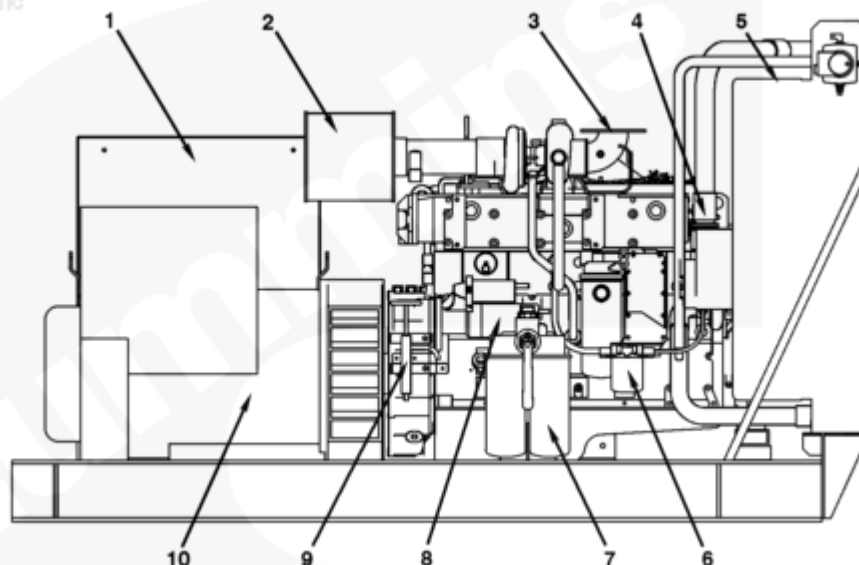
1. Local control panel
2. Air cleaner
3. Exhaust connection
4. Coolant thermostat location
5. Coolant expansion tank
6. Coolant filter
7. Starter
8. Duplex lubricating oil filters
9. Lubricating oil sump hand pump
10. Alternator.



00900426

Left View - C8.3 - C Power Keel Cooler Cooled

1. Aftercooler
2. Governor control
3. Emergency shutdown sensors
4. Duplex fuel filter
5. Lift pump
6. Oil fill
7. Fuel filters
8. Fuel pump.

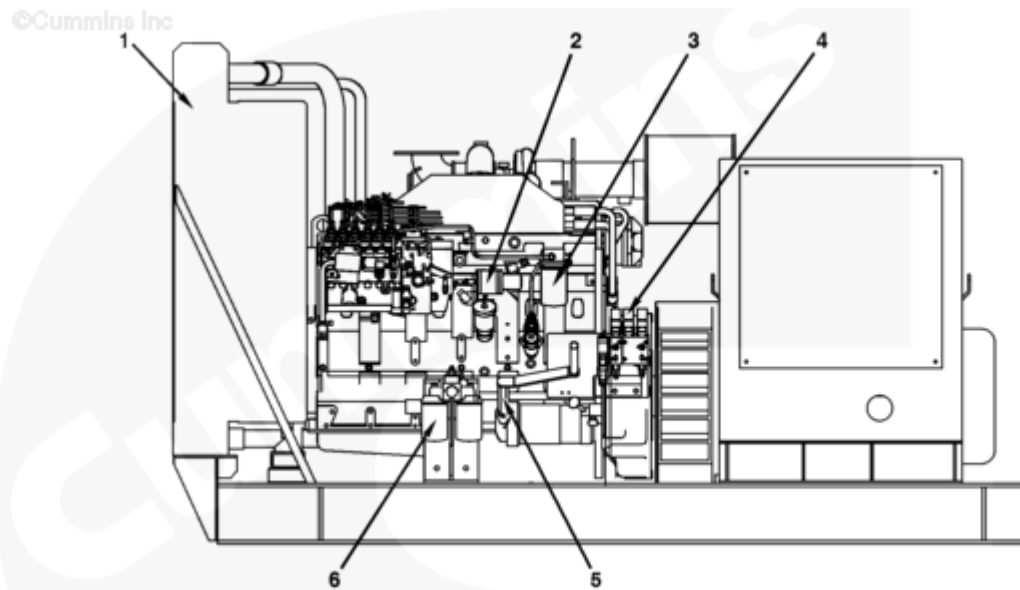


00900427

Right View - C 8.3 - C Safe Radiator Cooled

1. Local control panel
2. Air cleaner
3. Exhaust cConnection
4. Coolant thermostat location

5. Radiator fan drive shroud
6. Coolant filter
7. Duplex lubricating oil filter
8. Starter
9. Lubricating oil sump hand pump
10. Alternator.



Left View - C8.3 - C Safe Radiator Cooled

1. Radiator and expansion tank
2. Governor control
3. Fuel filters
4. Emergency shutdown sensors
5. Sight glass
6. Duplex fuel filters.